

Aadithyaa R

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LeetCode | GitHub | Portfolio | LinkedIn

SUMMARY

Budding Software Development Engineer passionate about building distributed solutions. Enjoys working across diverse technologies, continuously learning, and collaborating on meaningful projects.

EDUCATION

Shiv Nadar University

B.Tech Computer Science (Internet of Things) || CGPA – 8.65*

Player, Shiv Nadar University Table Tennis Team

Chennai, India

Aug 2023 – May 2027

Maharishi Vidya Mandir

Senior Secondary Education (Class XII) || Percentage – 94.60%

Hosur, India

Graduated: May 2023

PROJECTS

QEMULAB – Virtual Network Topology Builder | React, Node.js, Docker, QEMU, Guacamole

[GitHub](#)

- Engineered a browser-based virtualization lab orchestrating **KVM-accelerated QEMU** instances to simulate complex network topologies with real-time device interaction.
- Enabled clientless remote access by integrating **Apache Guacamole** with a custom **PostgreSQL** backend, delivering low-latency **VNC/Telnet** sessions directly within the web interface.
- Architected a custom networking layer using **multicast sockets** to simulate Layer 2 connectivity, allowing users to dynamically create isolated network segments.
- Containerized the microservices architecture using **Docker Compose**, optimizing storage efficiency via **QCOW2** copy-on-write overlays.

NetVerse Labs – Educational Metaverse | Next.js 16, TypeScript, Firebase, Gemini AI

[GitHub](#)

- Developed an immersive 2D virtual campus using **Next.js 16** and **TypeScript**, featuring a custom game engine built with **Framer Motion** for smooth character interaction.
- Embedded **Google Gemini AI** to power intelligent NPC tutors and integrated **Monaco Editor** for real-time in-game coding exercises.
- Orchestrated real-time multiplayer states and Role-Based Access Control (RBAC) using **Firebase**, synchronizing user presence and distinct teacher/student dashboards.
- Crafted a retro-futuristic "NetVerse OS" interface with **Shadcn UI** and integrated **Jitsi** video conferencing for virtual classrooms. [Live Demo](#).

ThinkBoard – Notes Web App | React, Express.js, DaisyUI, Upstash, MongoDB

[GitHub](#)

- Built a full-stack note-taking app with **React** and **Express-MongoDB**, styling the interface with **DaisyUI** and **Tailwind CSS**.
- Established secure CRUD operations with robust data validation and seamless note management functionality.
- Incorporated **Upstash Redis** for API rate limiting to prevent abuse and ensure service stability.
- Deployed on **Render**. [Live Demo](#).

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, Golang

Frameworks: React.js, Next.js, Node.js, Express.js, Gin, EJS

Tools: Git, VS Code, Docker, QEMU, Firebase

Libraries: Framer Motion, Shadcn UI, Apache Guacamole, Tailwind CSS, DaisyUI, Zustand

Databases: MongoDB, PostgreSQL, Redis, Firestore, Prisma(ORM)

OS: Linux, Windows

CERTIFICATIONS

- DBMS - SQL AND NoSQL
- DBMS - with JDBC
- Data Structures Using Java
- Programming Using Java
- Agile Scrum in Practice
- Software Engineering and Agile